FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – X
PAPER –2
TEST DATE: 21-05-2023

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 – 04, 19 – 22, 37 – 40)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A *(05 –10, 23 – 28, 41 – 46)*: This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B *(11 – 18, 29 – 36, 47 – 54)*: This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.**

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

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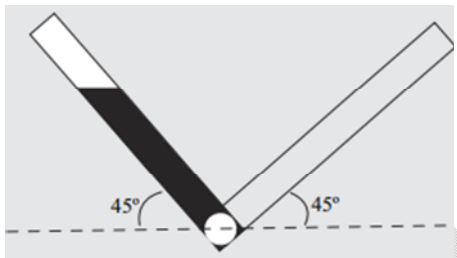
Physics

PART – I

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. A thin V-shaped glass tube is bent through the right angle and is fixed in the vertical plane as shown. Initially, the left part of the tube contains a column of water of length d. A valve at the bottom of the tube prevents the water from moving. At some instant, the valve is quickly opened. Neglecting friction, The time it takes the water to move completely into the right part of the tube is



(A) $\frac{\pi}{2}\sqrt{\frac{d}{\sqrt{2}g}}$ (B) $\pi\sqrt{\frac{d}{\sqrt{2}g}}$

(C) $2\pi\sqrt{\frac{d}{\sqrt{2}g}}$ (D) $4\pi\sqrt{\frac{d}{\sqrt{2}g}}$

2. A student of height h jumps vertically up from the "squat" position. At the top point of the jump, the student's center of mass is at a height 3h/4 from the ground. Find the average force F acting on the floor prior to the moment when the student loses contact with the floor. It is known that when the student stands on the floor, the center of mass is at a height h/2 from the floor; in the "squat" position, the center of mass is at a height h/4 from the floor. The mass of the student is m.

(A) 1 mg (B) 2 mg

(C) 3 mg (D) 4 mg

3. Consider the following nuclear reaction

$X^{200} \rightarrow A^{110} + B^{90} + \text{Energy}$

If the binding energy per nucleon for X, A and B are 7.4 MeV, 8.2 MeV and 8.4 MeV respectively, then the energy released is

(A) 100 MeV (B) 178 MeV

(C) 198 MeV (D) 220 MeV

4. Two concentric metal spheres of radii 'a' and 'b' (b > a) are separated by a medium that has dielectric constant ε_r and conductivity 'σ'. At time t = 0 an electric charge 'q_0' is suddenly placed on the inner sphere. What is the total current flowing through the medium at time 't'.

(A) $\frac{\sigma q_0}{\varepsilon_r \varepsilon_0} e^{-\frac{\sigma}{\varepsilon_r \varepsilon_0} t}$ (B) $\frac{\sigma q_0}{2\varepsilon_r \varepsilon_0} e^{-\frac{\sigma}{\varepsilon_r \varepsilon_0} t}$

(C) $\frac{\sigma q_0}{4\varepsilon_r \varepsilon_0} e^{-\frac{\sigma}{\varepsilon_r \varepsilon_0} t}$ (D) $\frac{2\sigma q_0}{\varepsilon_r \varepsilon_0} e^{-\frac{\sigma}{\varepsilon_r \varepsilon_0} t}$

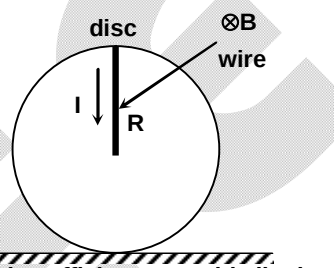
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Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. Potential at a point A is 3 volt and at a point B is 7 volt, an electron is moving towards A from B, then choose the correct option(s).
(A) It must have some K.E. at B to reach A
(B) It does not need to have any K.E. at B to reach A
(C) To reach A it must have more than or equal to 4 eV KE at B
(D) When it will reach A, it will have K.E. more than or at least equal to 4 eV if it was released from rest at B

6. A current carrying thin massless wire of length R is clamped along the radius of a uniform disc of radius R as shown in figure. A uniform magnetic field 'B' exists in the region. If a constant current I flows in the wire and there is no slipping between the disc and the horizontal surface, then choose the correct option(s).

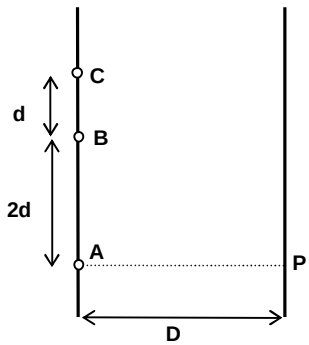


Friction is sufficient to avoid slipping

(A) Initial acceleration of the disc is $\frac{BIR}{m}$
(B) Initial friction acting on the disc is zero
(C) Maximum angular velocity of the disc would be $\sqrt{\frac{2BI}{3m}\left(\frac{\pi}{2}+2\right)}$
(D) Maximum angular velocity of disc would be $\sqrt{\frac{BI}{m}\left(\frac{\pi}{2}+1\right)}$

7. A sinusoidal input voltage source V = (2 + 3sinωt + 4cosωt) volt is connected across a load. Then choose the correct option(s).
(A) RMS value of the input voltage is $V_{rms} = \sqrt{\frac{33}{2}}$ volt
(B) RMS value of the input voltage is $V_{rms} = \left(\frac{2\sqrt{2}+7}{\sqrt{2}}\right)$ volt
(C) Average value of the input voltage is zero
(D) Average value of the input voltage is V_{av} = 2 volt

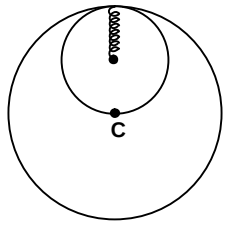
8. In the figure A, B and C are three slits each of them individually producing the same intensity I_0 at P when they are illuminated by parallel beam of light of wavelength λ. It is given that $BP - AP = \frac{\lambda}{2}$. Also given that d <<D, then wavelength λ and resultant intensity I at P will be (λ <<d).



(A) $\lambda = \frac{2d^2}{D}$
(B) $\lambda = \frac{4d^2}{D}$
(C) $I = 2I_0$
(D) $I = I_0$

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9. A non-conducting sphere having a spherical cavity centred at a distance 'a' from centre of sphere. Charge inside the sphere is distributed uniformly with volume density ρ. A particle of mass m having charge $-q$ is suspended vertically from light non-conducting spring of force constant k. Choose the correct statement(s). (there is no gravity and initially when $-q$ is suspended, the spring was in natural length)



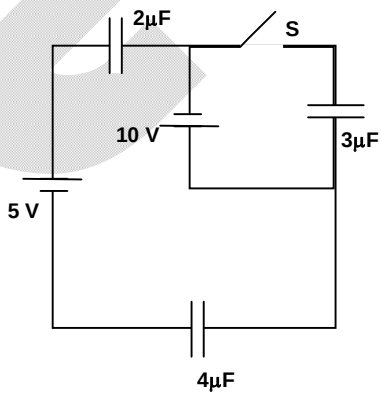
(A) Maximum elongation of the spring is $\left(\frac{2\rho aq}{3\varepsilon_0 k}\right)$

(B) Maximum elongation of the spring is $\left(\frac{\rho aq}{3\varepsilon_0 k}\right)$

(C) Time period of oscillation is $2\pi\sqrt{\frac{m}{k}}$

(D) Time period of oscillation is $2\pi\sqrt{\frac{\rho m}{\varepsilon_0 k}}$

10. In the given circuit initially switch 'S' is opened for a long time and then switch 'S' is closed. Then choose the correct option(s).



(A) The total energy stored by the capacitors before having switch 'S' closed is 150 μJ

(B) The total energy stored by the capacitors before having switch 'S' closed is $\frac{150}{9}$ μJ

(C) The total amount of heat generated is 150 μJ after the switch 'S' is closed.

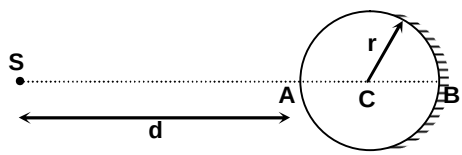
(D) The charge stored by 3 μF capacitor is 30 μC after the switch 'S' is closed.

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

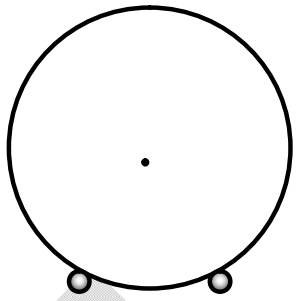
11. A balloon is filled with helium at the atmospheric pressure P. The volume of the balloon is V. The balloon is made of the material of mass m and density d. After being released, the balloon bursts at an altitude where the atmospheric pressure is (P/2). Immediately before bursting, the balloon has a volume of 1.25V. The maximum stress that the balloon material can withstand is $\sigma = \frac{kPVd}{16m}$. Find the value of k. (Assume that the temperature of helium remains constant, the balloon remains spherical, and the density of the material remains virtually constant)

12. A very expensive diamond is polished into a perfect sphere of radius r = 1 cm. The back surface of the sphere is then covered with silver. At what distance (in cm) in front of the sphere should a small source of light S be placed in order for the image to coincide with the source? (The index of refraction of diamond is μ = 2.4)

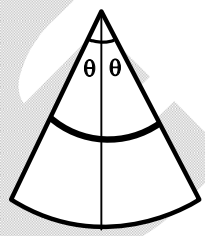


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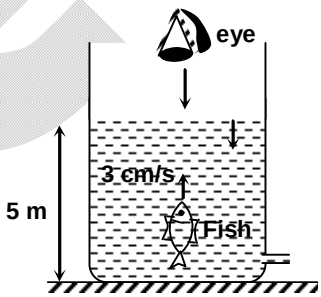
13. A sphere of radius r is rolling horizontally without slipping on two parallel rails placed a distance r/2 apart (in the diagram the sphere is rolling perpendicular to the plane of the page). If the centre of mass of the sphere has a speed v, the maximum velocity among all points of the sphere is $v_{max} = \left(\frac{3k + 4\sqrt{15}}{15}\right)v$. Find the value of k.



14. A rubber band has mass m and force constant k. When relaxed, the band forms a ring of radius r. The band is placed horizontally on a vertical frictionless cone as shown (angle 2θ is given). The radius R of the ring formed by the band is $R = r + \frac{mg}{\alpha\pi^2 k}\cot\theta$. Find the value of α.

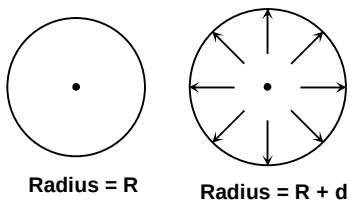


15. A tank having cross sectional area A has a hole at the bottom of area of cross section A_1 = A/1000. Bottom of the tank is a plane mirror. The tank contains water of refractive index 4/3. At the instant, when height of the water in the tank is 5m, a fish is rising vertically in the tank with a velocity 3 cm/sec toward the surface. Find the velocity of the fish as observed by the observer looking directly at the fish (in cm/s). (Take g = 10 m/s^2)



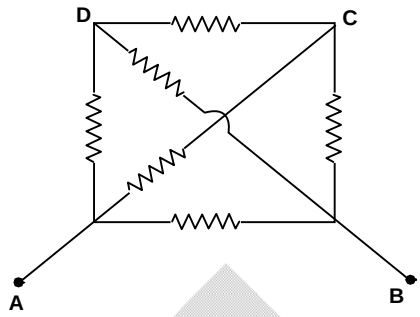
16. In a park there are three concentric circular running tracks. Radius of 2nd track is double the radius of first track and the radius of 3rd track is triple the radius of first track. Three runners are running on these tracks with constant speeds. When the runner in the first track completes one round, the runner in 2nd has completed half round and the runner in third track has completed quarter round. If the accelerations of the runners are in ratio $\alpha : \beta : \gamma$ where α, β and γ are least integers, then find the value of $\left(\frac{\alpha+\beta+\gamma}{3}\right)$.

17. A uniform solid ring of mass m and radius R is made of an elastic material. The symmetric radially outward force acting per unit length on the ring is F as shown by arrows in the figure. To increase the radius of the ring from R to (R + d), the magnitude of F required is $\frac{kAdY}{3R^2}$. Find the value of k. (Young's modulus of the material of ring is Y and area of its cross section is A)



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18. In the figure shown, each branch has resistance R = 2Ω, find the equivalent resistance (in Ω) between the points A and B.



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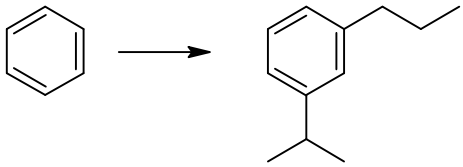
Chemistry

PART – II

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

19. Correct sequence for the following conversion, is



(A) (i) Friedal-Craft's alkylation (ii) reduction (iii) Friedal-Craft's alkylation
(B) (i) Friedal-Craft's acylation, (ii) reduction (iii) Friedal-Craft's alkylation
(C) (i) Friedal-Craft's alkylation, (ii) Friedal-Craft's acylation (iii) reduction
(D) (i) Friedal-Craft's acylation (ii) Friedal-Craft's alkylation (iii) reduction

20. For the reaction, $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$; $\Delta H^o = -802.8$ kJ

The value of $\Delta H_{C=0}$ from the following bond energy data will be

$\Delta H_{C\text{-}H} = 416.2$ kJ / mol

$\Delta H_{O=O} = 493.7$ kJ / mol

$\Delta H_{O\text{-}H} = 464.4$ kJ / mol

(A) 1263.1 kJ/mol (B) 798.7 kJ/mol
(C) 1292.4 kJ/mol (D) 987.8 kJ/mol

21. Among the oxides of nitrogen, the compound(s) in which of the following N atom is not directly attached with N atom

(A) N_2O (B) N_2O_3
(C) N_2O_4 (D) N_2O_5

22. Consider the following two equilibria at temperature T.

(i) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ $K_p = 4 \times 10^{-2}$ atm

(ii) $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$ $K_p = 2.0$ atm

Solid carbon, CaO, $CaCO_3$ are mixed and allowed to attain equilibrium at temperature T. The partial pressure of CO, when equilibrium is achieved, is

(A) 0.08 atm (B) 2.8 atm
(C) 0.56 atm (D) 0.28 atm

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

23. Which of the following salts react with dil. H_2SO_4 and liberate gas?

(A) $CaCO_3$ (B) KNO_2
(C) Na_2S (D) $ZnSO_3$

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24. The ammonia evolved from 0.5 g of the organic compound in Kjeldahl's estimation of nitrogen neutralizes 10 ml of 1 M H_2SO_4. Identify the correct statement(s) out of the following:
(A) Percentage of nitrogen in the organic compound is 56%
(B) Milliequivalents of ammonia produced in the solution are 20
(C) If the evolved ammonia were neutralized by 10 ml of 1 M HCl, the % of nitrogen would have been 28%
(D) Percentage of NH_3 in sample is 68%

25. Identify the correct statements:
(A) Λ_m increases with increase in temperature
(B) Λ_m decreases with increase in concentration
(C) Specific conductance increases with increase in concentration
(D) Specific conductance decreases with increase in temperature

26. Identify correct statement regarding gas (X).
NH_4^+ salt $+ OH^- \longrightarrow X(\text{gas})$
(A) AgI is soluble in (X)
(B) (X) turns red litmus blue and produces dense white fume with HCl
(C) (X) turns filter paper moistened with mercurous nitrate black and give intense blue colour solution with $CuSO_4$
(D) (X) when passed through Nessler's reagent produces Brown colour ppt.

27. Which of the following is correct about Cu^{+2}.
(A) Cu^{+2} form soluble complex with KI
(B) Iron and zinc when placed in Cu^{+2} salt solution, copper metal is formed.
(C) Cu^{+2} salts form soluble complex with excess KCN
(D) Cu^{+2} salts form soluble complex with NH_3

28. Find incorrect statement
(A) Depressant NaCN is used to separate ZnS and PbS ores in Froth Floatation process.
(B) NaCN selectively prevents PbS to come to the Froth but allows ZnS to come with the Froth
(C) Ni and Ti is purified by vapour phase refining
(D) Fe, Cu and Hg are reduced by self reduction process

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

29. How many salts from given below form precipitate of hydroxide and also soluble in excess of NaOH
$FeCl_3$, $HgCl_2$, $ZnSO_4$, $MnSO_4$, $AlCl_3$, $SnCl_4$

30. How many given below metal nitrates on heating gives metal oxide and NO_2 gas
Sr, Ba, Na, Ag, Zn, Rb, Li

31. Find the pH of solution, if 0.1 M CH_3COOH freezes at – 0.205°C if (K_f of water is 1.86 kg/mole/K).

32. 'x' moles of B_2H_6 reacts completely with CH_3OH and 6 moles of Boron containing product is formed. Find the value of 'x'

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33. An electron belongs to an orbit of H-atom whose angular momentum is 4.2178×10^{-34} kg – m^2 / sec. Find the number of visible line emitted when electron falls from this energy level to ground state.

34. How many of the following substance are more acidic than phenol?
O-Cresol, P-Cresol, Ethyl Alcohol, 2,4-dimethyl phenol, m-methoxy phenol, p-ethyl phenol, m-amino phenol.

35. $[A] + \text{Coke} \longrightarrow [B] \xrightarrow{H_3O^+} [C] \xrightarrow{\text{Red hot Cu–Tube}} \text{Mesitylene}$
The number of valence electrons in element [A] are

36. How many compounds are red in colour
HgI_2, HgO, Cu_2O, Pb_3O_4, Bi_2S_3, $PbCrO_4$, CdS

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Mathematics

PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

37. If x and y are real numbers such that $x \in [0, 4]$ and $y \in [0, 4]$. The probability that selected numbers satisfying $y^2 \le x$ is

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{1}{4}$ (D) $\frac{1}{16}$

38. For positive integers k = 1, 2, 3, ..., n. Let S_k denotes the area of ΔAOB_k (where O is origin) such that $\angle AOB_k = \frac{k\pi}{2n}$, OA = 1 and OB_k = k. Then the value of $\lim\limits_{n\to\infty} \frac{1}{n^2}\left(\sum\limits_{k=1}^{n} S_k\right)$ is

(A) $\frac{2}{\pi^2}$ (B) $\frac{4}{\pi^2}$

(C) $\frac{8}{\pi^2}$ (D) $\frac{1}{2\pi^2}$

39. Let A = $[a_{ij}]$ be a 3×3 matrix, where $a_{ij} = \lim\limits_{x\to 0} \frac{(1+ix)^{\frac{1}{j}} - 1}{x}$ for all $1 \le i, j \le 3$. If $A^2 = kA$, then k is

(A) 1 (B) –1

(C) 2 (D) 3

40. If a twice differentiable function satisfying a relation $f\left(x^2y\right) = x^2 f(y) + yf\left(x^2\right)$ $\forall$ x, y > 0 and f′(1) = 1, then the value of $f''\left(\frac{1}{7}\right)$ is

(A) $\frac{1}{7}$ (B) 7

(C) 3 (D) $\frac{2}{7}$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

41. L_1 is a tangent drawn to the curve $x^2 - 4y^2 = 16$ at $A\left(5, \frac{3}{2}\right)$. L_2 is another tangent parallel to L_1 which meets the curve at B. L_3 and L_4 are normals to the curve at A and B. Lines L_1, L_2, L_3, L_4 forms a rectangle, then

(A) equation of tangent at B is 6y = 5x – 16

(B) equation of normal at B is 12x + 10y + 75 = 0

(C) radius of largest circle inscribed in the rectangle is $\frac{32}{\sqrt{61}}$

(D) radius of the circle circumscribing the rectangle is $\frac{\sqrt{109}}{2}$

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42. Let P be any point on ellipse $3x^2 + 4y^2 = 12$ and S, S′ are its foci, then the locus of the centroid of triangle PSS′ is a conic C whose
(A) length of latus rectum equals 1
(B) locus of point of intersection of perpendicular tangents is $x^2 + y^2 = \frac{7}{9}$
(C) equation of auxiliary circle is $x^2 + y^2 = \frac{4}{9}$
(D) area of quadrilateral formed by tangents at the extremities of latus–rectum equals $\frac{16}{3}$

43. Consider the parabola whose equation is $y = x^2 - 4x$ and the line $y = 2x - b$. Then which of the following is/are CORRECT?
(A) For b = 9 the line is a tangent to the parabola
(B) For b = 7 the line cuts the parabola in A and B such that the $\angle AOB$ is a right angle where O is the origin
(C) For some $b \in R$ the line cuts the parabola in C and D such that x-axis bisects the $\angle COD$
(D) For b > 9 the line passes outside the parabola

44. Let z be a variable complex number satisfying $\frac{2z - \bar{z}}{3} = a - \frac{1}{3} + ib$ where $a, b \in R$ and $a^2 - b^2 = 1$.
If locus of z is the curve C, then
(A) director circle of the curve C is $(x + 1)^2 + y^2 = 10$
(B) area of the triangle formed by the straight line $x - 3y + 1 = 0$ and tangent at one of the vertices and the x-axis is $\frac{3}{2}$ sq. units
(C) distance between directrices of the curve C is $\frac{18}{\sqrt{10}}$
(D) equation of the circle passing through the points (2, 1) and both foci of the curve C is $x^2 + y^2 + 2x - 9 = 0$

45. Consider the binomial expansion of $\left(\sqrt{x} + \frac{1}{2 \cdot \sqrt[4]{x}}\right)^n$, $n \in N$ where the terms of the expansion are written in decreasing powers of x. If the coefficients of the first three terms form an arithmetic progression, then the statement(s) which hold good is/are
(A) total number of terms in the expansion of the binomial is 8
(B) number of terms in the expansion with integral power of x is 3
(C) there is no term in the expansion which is independent of x
(D) fourth and fifth are the middle terms of the expansion

46. Let lines normal to the circle $S_1 : x^2 + (y - 2)^2 = 4$ (with centre C_1) touches the circle $S_2 : (x - (r + 2))^2 + (y - 2)^2 = r^2$ (with centre C_2, $r > 0$) at A and B. If area of quadrilateral C_1AC_2B is 12 sq. units, then
(A) r = 3
(B) r = 6
(C) inradius of the triangle C_1AC_2 is 1
(D) Circumradius of the triangle C_1BC_2 is 5

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

47. If $\sin 24° = \sqrt{a} + \sqrt{b} - \sqrt{c - \sqrt{d}}$, where a, b, c, d ∈ N, then $\left(\frac{a + b + c + d}{12}\right)$ is

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48. If A_1, A_2, A_3, A_4 be the areas of the triangular faces of a tetrahedron, and h_1, h_2, h_3, h_4 be the corresponding altitudes of the tetrahedron. If volume of tetrahedron is 5 cubic units, then the minimum value of $\left|\frac{(A_1+A_2+A_3+A_4)(h_1+h_2+h_3+h_4)}{120}\right|$ is

49. It a plane is passing through three points A(a, 0, 0), B(0, b, 0), C(0, 0, c) with a > 0, b > 0, c > 0. If d be the distance between the origin O and the plane and m be the distance between the origin O and the point M(a, b, c), then minimum value of $\left(\frac{m}{d}\right)^2$ is

50. Sum of the series $\frac{3}{(1.2)^2}+\frac{5}{(2.3)^2}+\frac{7}{(3.4)^2}+\frac{9}{(4.5)^2}+.....\infty$ is

51. If $\lambda\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = (a+b+c)(b+c-a)(c+a-b)(a+b-c)$ where $a^2 = (x_1-x_2)^2+(y_1-y_2)^2$; $b^2 = (x_2-x_3)^2+(y_2-y_3)^2$; $c^2 = (x_3-x_1)^2+(y_3-y_1)^2$, then λ is

52. In an election, the number of candidates is one greater than the persons to be elected. If a voter can vote in 254 ways, by selecting atleast one of them, then the number of candidates is

53. If $I=\int\left(x^2+1\right)\left((x+1)e^x\right)^2 dx = A(f(x))^2 + c$, where c is constant of integration and $f(-1) = \frac{2}{e}$, then 2A + f(0) is

54. If $\left(1+x+x^2\right)^8 = \sum_{k=0}^{16} a_k x^k$, then the value of determinant $\begin{vmatrix} a_3 & a_7 & a_{13} \\ a_4 & a_6 & a_{12} \\ a_5 & a_4 & a_{11} \end{vmatrix}$ is

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